The Fan That Adapts

A 1200 mm BLDC ceiling fan whose **blades reshape themselves with room temperature** — no sensor, wire or battery on the rotor — with a **climate-aware speed controller** and a **tiny offline learner** built around them.

0
electronics on the rotor — the blade itself is the thermostat

+10–15 %
air at the same RPM when the room is hot (target, to be measured A/B)

2 nights
for the fan to learn a user's comfort offset; < 2 KB RAM, no cloud

≈ ₹15 k
Level-1 prototype; ≈ ₹350–500 added per unit at volume (est.)

A Team & institutional context

Team name	**[TEAM NAME]**
Institute	[Institute name, City]
Department / programme	[Department / Programme]
Faculty mentor	[Faculty mentor name, designation — or remove row]

Member	Programme & year	Role in this project	Email	Phone
Girish N	[Programme, Year]	Lead · firmware & learning	[email]	[phone]
[Member 2]	[Programme, Year]	Blade mechanics & aero tests	[email]	[phone]
[Member 3]	[Programme, Year]	Electronics & motor control	[email]	[phone]
[Member 4]	[Programme, Year]	Test rig, data & comfort trials	[email]	[phone]

B Track & areas of exploration

Track 1 · Fan. Primary area: **blade design for energy efficiency, air throw and low noise.** Combined with: **fan as a smart-home enabler (ambient temperature & humidity triggers).**

Why both: a blade that changes shape with heat is only half a product. The motor controller must know the room's temperature and humidity — and know what the blade is already doing — or the two will over-react to the same heat. The same sensors then make the fan the room's climate node for the AC and exhaust fan.

· How we got here

Before choosing a solution we mapped where ceiling fans under-perform in Indian homes: **28 room/usage conditions** (false ceilings, hot roofs, monsoon humidity, beds and mosquito nets under the fan, inverter power, …), each researched against ~600 sources, scored on a fixed rubric and stress-tested in a hostile review. The strongest finding drove everything in this abstract:

> **Fans are designed and rated at maximum speed in an open test chamber. They are used at low speed, at night, over a bed, while the room cools several °C and humidity climbs.** A fixed blade and a fixed speed step cannot be right for both a 34 °C afternoon and a 25 °C dawn — so the user wakes up and reaches for the remote. We found no fan product or active patent that changes blade shape automatically with the room.

Reading guide — p.1: idea and quantified claims · p.2: why it is technically feasible, including safety · p.3: components, cost, sourcing risk and the build plan to Round 2.

1 The idea — what we are building and how it works

We are building a ceiling fan that adapts at three levels. **Body:** the outer 35 % of each blade carries a 25 %-chord trailing-edge flap turned by a helical **thermostatic-bimetal coil** that sits in the blade's own airflow. Above a 28 °C neutral point the flap deflects down (~0.8° per °C), adding camber and effective pitch so every revolution moves more air; below it the flap rises and the fan gives a softer, lower-power draft. Nothing on the rotor needs power — the blade is the thermostat, and it keeps working on a plain wall regulator. **Reflex:** a temperature/humidity sensor feeds firmware that converts T × RH into a target air speed at the person (ASHRAE-55 / Indian IMAC comfort models), then shapes RPM into a slow, natural "breeze", an inaudible sleep ramp, or a humid-flush pattern. **Brain:** an ESP32-C6 learns each household's comfort offset from their own remote overrides and shifts the Reflex target — fully offline, < 2 KB of RAM. The firmware infers the blade's current shape from motor current, so the mechanics and the software never double-react to the same heat.

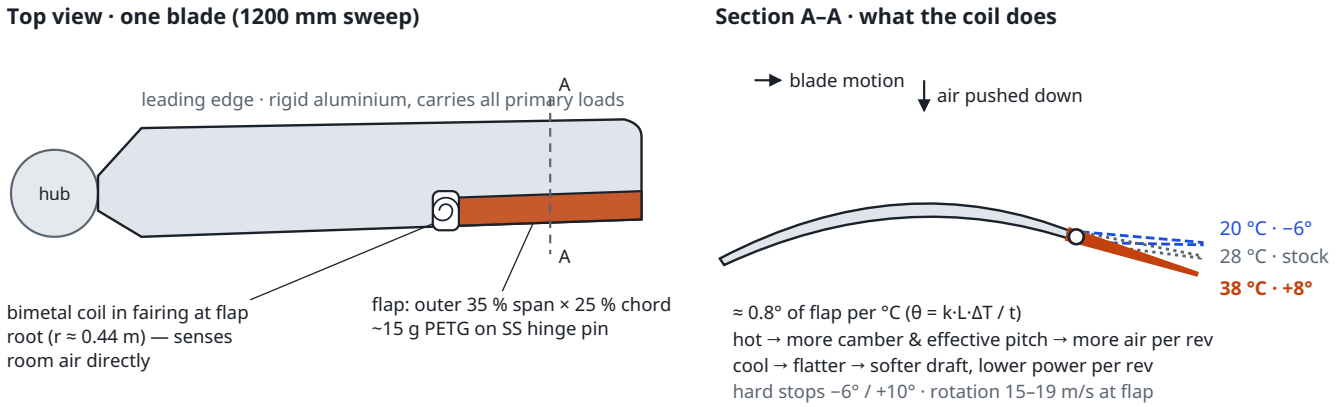


Fig. 1 — The Body layer. The rigid front 75 % of the blade is unchanged and carries every primary load; only a light flap moves. Coil length 250 mm, strip 20 × 0.5 mm, high-flexivity bimetal.

2 Baseline vs target — what we will prove, and how

Metric	Baseline (today's fan)	Our target	How we measure it
Air delivery at a fixed speed setting, room at 35 °C vs 22 °C	0 % — blade is fixed	**+10 to +15 % hot** **−8 to −10 % cool**	Same motor & RPM, swap stock ↔ morphing blades (true A/B); anemometer grid 1.5 m below
Power at a fixed speed setting in a cool room (20–22 °C)	unchanged	**≈ −20 to −25 %**	Wattmeter on driver input
Energy / electronics needed for that adaptation	—	**0 W · 0 sensors on rotor**	By design; demo with electronics switched off
Flap response to a room temperature step	—	**τ ≈ 10–30 s · ≥ 0.6°/°C**	Thermal chamber 15 → 45 °C, laser-on-mirror
Mean RPM needed for the same comfort vote (breeze vs steady air)	steady RPM	**≥ 10 % lower RPM (≈ −25 % motor power)**	Blind A/B, ≥ 8 volunteers, sensation vote at 30 °C
Night: speed-change rate & noise at bed height	step change per speed level	**≤ 3 RPM/s ramps · ≤ 32 dB(A)**	Sound meter 1.2 m below hub; RPM log from tach
Manual remote corrections per night	logged in week 1 (3 users)	**≤ 0.3 after 7 nights**	30-day synthetic user, then live logs
AC set-point at equal comfort (stretch)	24 °C	**26 °C (≈ −12 % AC energy)**	BEE: ~6 % AC energy per °C

3 Why we believe this is new

- **A passively morphing ceiling-fan blade.** Passive thermal pitch control exists only in an expired 1982 automotive engine-fan patent (US 4,487,551); fan patents we found change pitch only manually. Full search before Round 2.
- **Motor current as a blade-shape sensor.** A morphed blade needs more torque at the same RPM; the driver's current-vs-RPM residual tells the firmware the flap angle for free — closing the loop between passive mechanics and active control.
- **Learning a comfort offset, not a schedule.** "Speed 4 at 11 pm" learned in a heatwave misfires in the monsoon; "this user likes 1 °C cooler than the model" does not.

4 Technical feasibility — how it works and why it is buildable

Body — the morphing blade (numbers we designed to)

Actuator choice. We compared eight smart materials. **Thermostatic bimetal** wins: linear, no functional hysteresis or fatigue, cheap, and proven for decades inside rotating automotive fan clutches. NiTi shape-memory wire with a steel bias spring is our backup (two-state "snap" version at the blade root); shape-memory polymers and liquid-crystal elastomers creep under centrifugal load, wax actuators weigh 50–120 g and take minutes, and "trained" two-way NiTi loses 30–60 % of its stroke over time.

Stroke. Coil rotation $\theta = k \cdot L \cdot \Delta T / t$: with specific curvature $k \approx 28 \times 10^{-6}\ K^{-1}$ (high-flex grade, to confirm on datasheet), L = 250 mm, t = 0.5 mm → **≈ 0.8° per °C**, ±8° over a ±10 °C swing. With flap effectiveness ≈ 0.5, that is ~4° effective angle change on the outer span — which, by r^2 weighting, produces ~60 % of the blade's thrust. Since flow scales roughly with √thrust, **+10–15 % air delivery** at the same RPM is a conservative blade-element estimate.

Loads. Dynamic pressure at the flap (r ≈ 0.52 m, 350 RPM) is ≈ 220 Pa, giving a hinge moment of ≈ 3 N·mm. Coil stiffness is ≈ 2 N·mm per degree (blocked torque ≈ 17 N·mm per 10 °C), and a 20 % overhang balance halves the air load → **speed-induced flap error < 1°**; the slight back-off at top speed acts as a natural limiter. Centrifugal pull on the 15 g flap is **10 N at 350 RPM and 16 N at the 1.25× overspeed test (438 RPM)**, taken axially by a thrust shoulder on a 3 mm stainless hinge pin rated well over 1 kN.

Thermal response. A 0.5 mm steel strip in 5–19 m/s airflow has a time constant of ≈ 10–30 s — the blade tracks the room, not the motor (the coil sits 0.44 m from the hub). Ceiling-level air is 1–2 °C warmer than at the bed; the neutral point is set by rotating the coil arbor (a calibration screw), not by changing parts.

Reflex — climate-tuned speed patterns

Air movement of ~1 m/s at the occupant offsets ≈ 2–3 °C of operative temperature (ASHRAE 55, SET method). A 16-row T × RH map (50–350 RPM) sets the base speed. Human studies show fluctuating airflow at **0.2–1 Hz with a 1/f ("natural wind") spectrum** feels cooler than steady air at the same mean speed. The rotor's inertia (≈ 0.06 kg·m²) makes 0.5 Hz speed swings costly (~+28 % energy), so we split the job: firmware makes **slow ±10–15 % swings over 10–30 s with jerk-limited S-curves (≤ 12 RPM/s, ≤ 10 % extra torque)**, and the blade wake supplies the fast gusts. **Sleep** holds a ≤ 150 RPM band and ramps at ≤ 3 RPM/s as the room cools. **Humid-flush** (≥ 32 °C and ≥ 70 % RH, where extra speed adds little evaporative cooling) uses a low base speed with a short burst every ~45 s.

Brain — tiny offline learning

Overrides are binned by half-hour × {awake, asleep} into a table of comfort offsets (°C) with Beta-Bayesian confidence. A change is adopted only if it repeats on **two consecutive nights**; one-offs decay. It runs in **< 2 KB RAM and < 100 cycles per update** on an ESP32-C6 (≈ ₹113 at volume, Wi-Fi 6/BLE/Thread → Matter). We rejected on-device neural networks (forgetting, flash wear) and bandit "exploration" (nobody wants an experiment at 2 am). A 24 GHz radar sees a **sleeping** person's breathing where PIR fails; lights-off marks bedtime. Cold start uses the physics map; seasons shifting the weekly mean temperature fall back to physics until new habits form; two disagreeing users default to the lower speed. Learned data is saved daily plus a brown-out-triggered save (~12 ms); everything stays on the device.

> **Safety-critical aspects.** (1) **No electronics, wiring or slip rings on the rotor.** (2) Flap retained by pin + thrust shoulder designed for the 438 RPM overspeed case; hard stops at −6° / +10° prevent over-travel above 45 °C; a light torsion spring returns the flap to the stock (neutral) shape if a coil breaks — failure looks like a normal fan. (3) Three actuators matched as a set; blade-tip tracking ≤ 2 mm verified by strobe before every spin test; spin tests run behind a mesh guard. (4) Bimetal isolated from aluminium by POM bushes and nickel plating (monsoon galvanic corrosion). (5) A manual command always overrides automation within 50 ms; RPM limits are clamped in the driver; a watchdog falls back to the last manual speed. (6) Our prototype electronics are **24 V SELV** — no team-built mains circuitry; the product path is a UART link to the fan's existing controller.

Situations it handles — one product, not eight modes

Situation	What the fan does	Layer
Hot afternoon, 34 °C / 50 % RH	Flaps down (more air per rev) + breeze pattern at a higher target speed	Body + Reflex
Cool dawn or winter, 22–25 °C	Flaps up — softer, lower-power draft; base speed steps down without a click	Body + Reflex
Monsoon, 30 °C / 80 % RH	Humid-flush bursts instead of pointless top speed	Reflex
Person asleep and still	Radar keeps Sleep mode on (a PIR fan would switch off)	Brain
User always wants it cooler around 1 am	Learns a −1 °C offset; applies it from night 3	Brain
Power cut, Wi-Fi down, plain regulator	Blades still adapt; learned habits survive in flash	Body
Electronics fault	Watchdog → last manual speed; blade adaptation unaffected	Safety

5 Execution feasibility

System-level block diagram

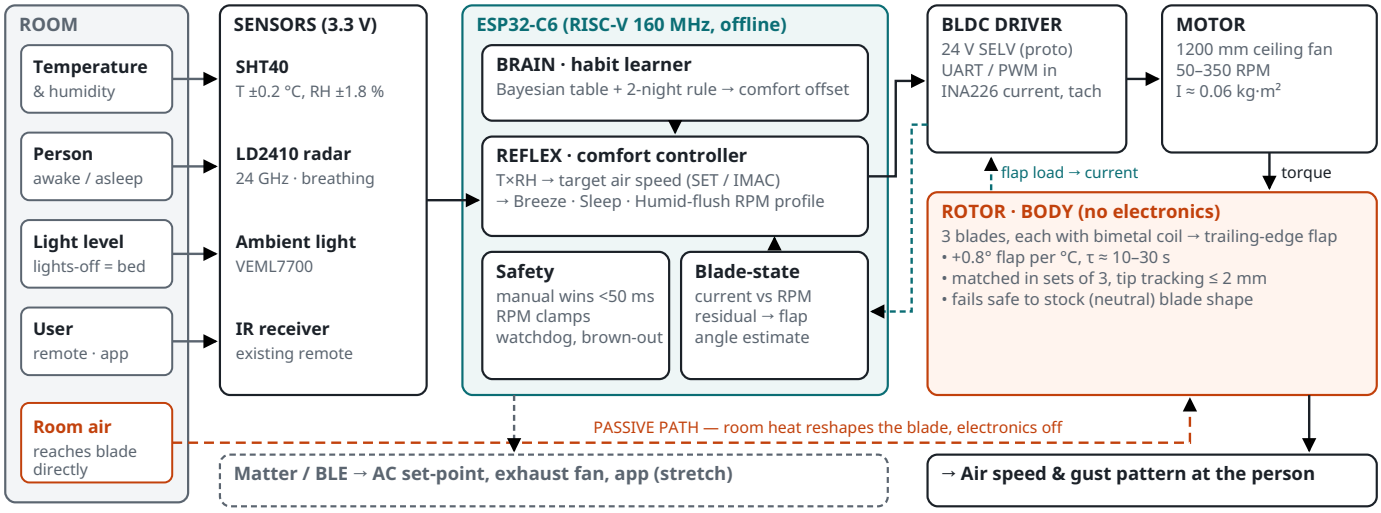


Fig. 2 — Two independent paths reach the air: an electronic path (sensors → ESP32-C6 → driver → motor) and a passive path (room air → bimetal → flap). Motor current closes the loop between them.

Components & rough BOM (Level-1 prototype)

Item	≈ ₹	Risk
1200 mm 24 V BLDC ceiling fan + 2nd blade set (stock A/B reference)	**4,500**	Med
High-flex bimetal strip, 1 m (coils wound in-house)	**1,200**	Med
NiTi wire + bias springs (Concept B backup)	**1,300**	Low
PETG flaps, SS pins, POM bushes, torsion springs	**1,500**	Low
ESP32-C6 DevKit, SHT40, LD2410, VEML7700, IR rx	**2,000**	Low
INA226, optical tach, 24 V SMPS, buck, wiring	**1,200**	Low
Vane anemometer, sound meter, thermal test box	**3,300**	Low
Total (order of magnitude)	**≈ 15,000**	

At volume (our estimate): 3 coils + flaps + pins ≈ ₹150–250; sensors + MCU ≈ ₹200–250, less if the fan's existing Wi-Fi module hosts the firmware.

Component sourcing risk

- **Bimetal strip** — Indian manufacturer (Shivalik Bimetal), but sample lead time may be 1–3 weeks. *Fallback:* coils from dial thermometers / automotive fan-clutch spares for the first static tests.
- **24 V BLDC fan** — must accept an external speed command. *Fallback:* off-the-shelf 24 V sensorless BLDC driver.
- **Custom-fabricated:** flaps (3D-printed), coil winding + stress relief, trailing-edge slot machining — all in-house.
- **Imported:** LD2410, ESP32-C6 modules — stocked by Indian distributors; nothing long-lead.

What we deliberately do not claim

Morphing does not give "free" air: by momentum theory +12 % flow costs ~+40 % ideal power whichever way you get it. The gain is **who decides** (the blade, passively), a **lower-power gentle mode** when cool, and — our hypothesis to test — **less noise** than buying the same air with tip speed (+12 % RPM ≈ +2.5 dB by V^5 scaling).

Build plan to Round 2 — a working Level-1 prototype

Week	Body (mechanics)	Reflex + Brain (firmware)
1	Order parts; wind & stress-relieve 4 coils; static chamber 15 → 45 °C. **Go/no-go:** ≥ 0.6°/°C, trio matched within ±0.1°/°C — else switch to NiTi root hinge (Concept B).	ESP32-C6 bring-up: SHT40, radar, IR remote decode, tach, INA226; closed-loop RPM control on the 24 V driver.
2	Slot blades, fit flaps; spin test 50 → 350 → 438 RPM behind guard; strobe tip tracking; record current-vs-RPM signature at each flap angle.	T × RH map + Breeze / Sleep / Humid-flush profiles; habit learner validated on a 30-day synthetic user on PC, then ported.
3	A/B airflow grid (stock vs morphing set) with the room at ~22 °C and heated to ~35 °C; power and noise at each speed.	Blade-state estimator from current residual; integrate all three layers; safety overrides & watchdog tests.
4	**Level-1 demo:** one fan, three layers. Warm the blade and watch the flap move with every electronic switched off; anemometer reading rises live. Then a live dashboard of T, RH, RPM, estimated flap angle and learned offset — plus our measured data pack against every target in Section 2.	